Geoffrey Lee

g42lee@uwaterloo.ca linkedin.com/in/lgeoff31 github.com/lgeoff31 geoffreylee.me

Education

University of Waterloo, Bachelor of Software Engineering Graduating April 2028

- **GPA:** 3.8/4.0
- **CS:** OS, Distributed Systems, Networking, Data Structures, Algorithms, Database Systems
- **Math:** Combinatorics, Optimization, Statistics, Linear Algebra, Numerical Computation

Employment

Atoms | Member of Technical Staff New York, New York | Incoming 2026

- Incoming Member of Technical Staff Intern

Biztrip AI | Software Engineer Intern San Francisco, CA | Sep 2025 – Dec 2025

- Built an automated fare-monitoring and rebooking system in **Python (FastAPI)** that polls flight-pricing data and auto-reissues tickets when fares drop, **saving $15K/month** in corporate travel spend while preserving itinerary constraints.
- Engineered a **proactive AI agent** using **TypeScript** to detect upcoming travel signals from calendar events, orchestrating flight and hotel bookings, reducing total corporate travel coordination by **1500+ hours/year**.

Shopify | Software Engineer Intern Toronto, ON | Jan 2025 – Apr 2025

- Integrated an autofill checkout option (Shop Pay) that pre-populates tokenized payment and address details from the Shop app profile, **reducing time-to-checkout by ∼90s** and increasing conversion by **4.8%**, built with **Rails and GraphQL**.
- Developed and ran **A/B tests** on an optional checkout step within the 4-step signup flow to highlight subscription-based plans, improving **early conversion rates** by **4%** and projecting **$8M+** in ARR, using **React** and **Typescript**.

RideCo | Software Engineer Intern Waterloo, ON | May 2024 – Aug 2024

- Built a **real-time vehicle tracking** platform using **Django and React** that ingests millions of live GPS updates/day with **high-performance** interpolation and debouncing algorithms, adopted by **100% of the dispatch operations team**.
- Integrated a ride replay system to visualize historical rides to streamline incident investigations and data analysis, reducing costs by **$300K** across **850 incidents** and **2300+ hours annually**.

Projects

Echo Cache

- Built a distributed caching system in **C++** designed for **horizontal scalability**, partitioning data across multiple worker nodes with automatic replication for **fault tolerance**.
- Designed a **multi-threaded orchestrator** that handles **concurrent** client connections over **TCP**, with failover reads to replica nodes and idle connection cleanup.

Bisect

- Built an interactive AI debugging platform that isolates the first **regression-introducing commit** by launching a user's repository at selected commits utilizing a **binary search algorithm**.
- Leveraged AI-driven code diff analysis to identify root-cause changes and auto-generate patch PRs, cutting **end-to-end debugging time by ∼30% per regression**.

Sync AI

- Built an AI-powered app for scheduling group activities based on common availability using **OpenAI, Google Places API, and Google Calendar**, reducing ∼30 minutes of planning to a **single click**.

Skills

- **Languages:** C++, C, Python, Java, Rust, TypeScript, JavaScript, Ruby, SQL, Shell Script
- **Technologies:** Docker, PostgreSQL, GraphQL, NodeJS, FastAPI, Rails, Django, React, NextJS, Git
- **Systems:** Distributed Systems, Network Programming, Concurrency, Memory Management, Fault Tolerance